United States Patent	12400972
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Tanioka; Tomonori et al.

Semiconductor device for suppressing excessive wetting and spreading of bonding layer

Abstract

A semiconductor device includes: a support member including a main surface facing a thickness direction; a semiconductor element mounted on the main surface; and a bonding layer interposed between the support member and the semiconductor element, wherein the support member is formed with a first protrusion that protrudes from the main surface, and wherein the first protrusion surrounds the semiconductor element when viewed in the thickness direction.

Inventors: **Tanioka; Tomonori (Kyoto, JP), Egami; Kazuo (Kyoto, JP)**

Applicant: **ROHM CO., LTD.** (Kyoto, JP)

Family ID: **1000008777479**

Assignee: **ROHM CO., LTD. (Kyoto, JP)**

Appl. No.: **17/843319**

Filed: **June 17, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230005845 A1	Jan. 05, 2023

Foreign Application Priority Data

JP	2021-109982	Jul. 01, 2021

Publication Classification

Int. Cl.: **H01L23/00** (20060101); **H01L23/31** (20060101); **H01L23/495** (20060101)

U.S. Cl.:

CPC **H01L23/562** (20130101); **H01L23/3107** (20130101); **H01L23/49513** (20130101); **H01L23/4952** (20130101); **H01L23/49524** (20130101); **H01L23/49548** (20130101);

Field of Classification Search

CPC: H01L (23/562); H01L (23/3107); H01L (23/49513)

USPC: 257/669

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4994897	12/1990	Golubic	257/667	H01L 23/4334
2021/0118815	12/2020	Kosugi	N/A	H01L 24/48

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2006303216	12/2005	JP	N/A
2017005149	12/2016	JP	N/A
2017-174951	12/2016	JP	N/A
2018163962	12/2017	JP	N/A
2011145176	12/2010	WO	N/A
2017169485	12/2016	WO	N/A

OTHER PUBLICATIONS

Office Action issued in Japanese Application No. 2021-109982, dated Mar. 18, 2025, with English translation. cited by applicant
Notice of Reasons for Refusal issued in Patent Application Number: Japanese Patent Application No. 2021-109982, mailed Jun. 10, 2025, with English translation. cited by applicant

---

Primary Examiner: Hoang; Quoc D

Attorney, Agent or Firm: HSML P. C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2021-109982, filed on Jul. 1, 2021, the entire contents of which are incorporated herein by reference.
TECHNICAL FIELD
(2) The present disclosure relates to a semiconductor device.
BACKGROUND
(3) In the related art, there is disclosed an example of a semiconductor device including a first lead including a first pad having a pad main surface, a semiconductor element mounted on the pad main

surface, and a sealing resin that is in contact with the pad main surface and covers the semiconductor element. The semiconductor element is conductively bonded to the first pad via a bonding layer. The semiconductor device further includes a second lead including a second pad and a wire (first bonding wire) conductively bonded to the semiconductor element and the second pad. The second pad and the wire are covered with the sealing resin. As a result, in the semiconductor device, the semiconductor element and members related to a conduction path of the semiconductor element are protected from external factors by the sealing resin.
(4) In the manufacture of the semiconductor device disclosed in the related art, in a case where the semiconductor element is bonded to the first pad via the bonding layer, when the bonding layer melted by reflow becomes excessively wet and spreads, a position deviation of the semiconductor element with respect to the pad main surface becomes relatively large. In this case, when the wire is conductively bonded to the semiconductor element in the manufacture of the semiconductor device, a length of the wire may become excessive. Further, the melted bonding layer may overcome the first pad to reach the second pad. In this case, a short-circuit occurs between the first lead and the second lead. Therefore, it is required to suppress excessive wetting and spreading of the bonding layer.
SUMMARY
(5) Some embodiments of the present disclosure provide a semiconductor device capable of suppressing excessive wetting and spreading of a bonding layer interposed between a support member and a semiconductor element.
(6) According to one embodiment of the present disclosure, there is provided a semiconductor device including: a support member including a main surface facing a thickness direction; a semiconductor element mounted on the main surface; and a bonding layer interposed between the support member and the semiconductor element, wherein the support member is formed with a first protrusion that protrudes from the main surface, and wherein the first protrusion surrounds the semiconductor element when viewed in the thickness direction.
(7) Other features and advantages of the present disclosure will become more apparent with the detailed description given below based on the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view of a semiconductor device according to a first embodiment of the present disclosure.
(2) FIG. **2** is a plan view of the semiconductor device shown in FIG. **1**.
(3) FIG. **3** is a plan view corresponding to FIG. **2**, in which a sealing resin is transparent.
(4) FIG. **4** is a bottom view of the semiconductor device shown in FIG. **1**.
(5) FIG. **5** is a front view of the semiconductor device shown in FIG. **1**.
(6) FIG. **6** is a cross-sectional view taken along line VI-VI of FIG. **3**.
(7) FIG. **7** is a cross-sectional view taken along line VII-VII of FIG. **3**.
(8) FIG. **8** is a partially enlarged view of the semiconductor element shown in FIG. **3** and its vicinity.
(9) FIG. **9** is a partially enlarged view of a pad portion of a support member shown in FIG. **6** and its vicinity.
(10) FIG. **10** is a partially enlarged view of a cover portion of a terminal (first terminal) shown in FIG. **6** and its vicinity.
(11) FIG. **11** is a partially enlarged plan view of a semiconductor element and its vicinity according to a modification of the semiconductor device shown in FIG. **1**, in which the sealing resin is transparent.

(12) FIG. **12** is a plan view of a semiconductor device according to a second embodiment of the present disclosure, in which a sealing resin is transparent.
(13) FIG. **13** is a cross-sectional view taken along line XIII-XIII of FIG. **12**.
(14) FIG. **14** is a partially enlarged view of a semiconductor element shown in FIG. **12** and its vicinity.
(15) FIG. **15** is a partially enlarged view of a pad portion of a support member shown in FIG. **13** and its vicinity.
(16) FIG. **16** is a partially enlarged view of a cover portion of a terminal (first terminal) shown in FIG. **13** and its vicinity.
(17) FIG. **17** is a partially enlarged cross-sectional view of a pad portion of a support member and its vicinity according to a modification of the semiconductor device shown in FIG. **12**.
DETAILED DESCRIPTION
(18) Embodiments for carrying out the present disclosure will now be described with reference to the accompanying drawings.
First Embodiment
(19) A semiconductor device A**10** according to a first embodiment of the present disclosure will be described with reference to FIGS. **1** to **10**. The semiconductor device A**10** is used in electronic apparatuses and the like including a power conversion circuit such as a DC-DC converter. The semiconductor device A**10** includes a support member **10**, a plurality of terminals **20**, a semiconductor element **30**, a first bonding layer **39**, a plurality of conductive members **40**, and a sealing resin **50**. Here, in FIGS. **3** and **8**, the sealing resin **50** is transparent for the sake of convenience of understanding. In FIG. **3**, the transparent sealing resin **50** is indicated by an imaginary line (two-dot chain line).
(20) In the description of the semiconductor device A**10**, for the sake of convenience, a thickness direction of the support member **10** is referred to as a “thickness direction z.” A direction orthogonal to the thickness direction z is referred to as a “first direction x.” A direction orthogonal to both the thickness direction z and the first direction x is referred to as a “second direction y.” When viewed in the thickness direction z, the first direction x corresponds to a longitudinal direction of the semiconductor device A**10**. When viewed in the thickness direction z, the second direction y corresponds to a lateral direction of the semiconductor device A**10**.
(21) As shown in FIGS. **3**, **6**, and **7**, the support member **10** mounts the semiconductor element **30** and forms a portion of a conduction path between the semiconductor element **30** and a wiring board on which the semiconductor device A**10** is mounted. The support member **10** includes a first base **101**. The first base **101** forms a main part of the support member **10**. The first base **101** is obtained from a lead frame containing copper (Cu) or a copper alloy. Therefore, the support member **10** has conductivity.
(22) As shown in FIGS. **3** and **7**, the support member **10** includes a pad portion **11** and a terminal portion **12**. In the semiconductor device A**10**, the pad portion **11** and the terminal portion **12** include the first base **101**. The pad portion **11** includes a first main surface **111**, a back surface **112**, and a through-hole **113**. A portion of the pad portion **11** is covered with the sealing resin **50**. The first main surface **111** faces one side of the thickness direction z and is opposed to the semiconductor element **30**. The first main surface **111** is in contact with the sealing resin **50**. The back surface **112** faces an opposite side of the first main surface **111** in the thickness direction z. The back surface **112** is plated with, for example, tin (Sn). The back surface **112** is exposed from the sealing resin **50**. The through-hole **113** extends from the first main surface **111** to the back surface **112** in the thickness direction z and penetrates the pad portion **11**. The through-hole **113** has a circular shape when viewed in the thickness direction z. As shown in FIG. **6**, a thickness T of the first base **101** included in the pad portion **11** is greater than the maximum thickness t.sub.max of a second base **201** (details thereof will be described later) included in each of the plurality of terminals **20**.

(23) As shown in FIGS. **3** and **7**, the terminal portion **12** includes a portion extending along the first direction x and is connected to the pad portion **11**. Therefore, the pad portion **11** and the terminal portion **12** are electrically connected to each other. A part of the terminal portion **12** is covered with the sealing resin **50**. The part of the terminal portion **12** covered with the sealing resin **50** is bent when viewed in the second direction y. A surface of a part of the terminal portion **12** exposed from the sealing resin **50** is plated with tin.
(24) As shown in FIGS. **6** to **9**, the first main surface **111** of the pad portion **11** includes a first region **111**A and a second region **111**B. In FIGS. **3** and **8**, the second region **111**B is shown by a region of a plurality of straight lines. The first region **111**A overlaps the semiconductor element **30** when viewed in the thickness direction z. The first region **111**A is a flat surface. The second region **111**B surrounds the first region **111**A. As shown in FIG. **9**, the pad portion **11** is formed with a plurality of grooves **16** that are recessed from the second region **111**B and are located apart from each other. The plurality of grooves **16** extend in a direction orthogonal to the thickness direction z. The sealing resin **50** enters the plurality of grooves **16**. The plurality of grooves **16** are formed by subjecting the second region **111**B to laser machining. As the plurality of grooves **16** are formed in the pad portion **11**, a surface roughness of the second region **111**B is greater than a surface roughness of the first region **111**A.
(25) As shown in FIGS. **3**, **6**, and **7**, the semiconductor element **30** is mounted on the first main surface **111** of the pad portion **11** of the support member **10**. In the semiconductor device A**10**, the semiconductor element **30** is an n-channel type MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor) of a vertical structure. The semiconductor element **30** includes a compound semiconductor substrate. The main material of the compound semiconductor substrate is silicon carbide (SiC). In addition, silicon (Si) may be used as the main material of the compound semiconductor substrate. In the semiconductor device A**10**, an area of the semiconductor element **30** is 40% or less of an area of the first main surface **111** when viewed in the thickness direction z. The semiconductor element **30** is not limited to the MOSFET. The semiconductor element **30** may be another transistor such as an IGBT (Insulated Gate Bipolar Transistor). Further, the semiconductor element **30** may be an LSI or a diode. The semiconductor element **30** includes a first electrode **31**, a second electrode **32**, and a third electrode **33**.
(26) As shown in FIGS. **8** and **9**, the first electrode **31** is provided on the side which the first main surface **111** of the pad portion **11** of the support member **10** faces in the thickness direction z. A current corresponding to power after being converted by the semiconductor element **30** flows through the first electrode **31**. That is, the first electrode **31** corresponds to a source of the semiconductor element **30**.
(27) As shown in FIG. **9**, the second electrode **32** is provided on an opposite side of the first electrode **31** in the thickness direction z. The second electrode **32** faces the first main surface **111** of the pad portion **11** of the support member **10**. A current corresponding to power before being converted by the semiconductor element **30** flows through the second electrode **32**. That is, the second electrode **32** corresponds to a drain of the semiconductor element **30**.
(28) As shown in FIG. **8**, the third electrode **33** is provided on the same side as the first electrode **31** in the thickness direction z and is located away from the first electrode **31**. A gate voltage for driving the semiconductor element **30** is applied to the third electrode **33**. That is, the third electrode **33** corresponds to a gate of the semiconductor element **30**. An area of the third electrode **33** is smaller than an area of the first electrode **31** when viewed in the thickness direction z.
(29) As shown in FIG. **9**, the first bonding layer **39** is interposed between the first main surface **111** of the pad portion **11** of the support member **10** and the second electrode **32** of the semiconductor element **30**. In the semiconductor device A**10**, the first bonding layer **39** is in contact with the first region **111**A of the first main surface **111**, and the second electrode **32**. The first bonding layer **39** contains a metal element. The metal element is, for example, tin. The first bonding layer **39** is solder. The second electrode **32** is conductively bonded to the pad portion **11** via the first bonding

layer **39**. Therefore, the terminal portion **12** of the support member **10** corresponds to a drain terminal of the semiconductor device A**10**.

(30) As shown in FIGS. **8** and **9**, the pad portion **11** of the support member **10** is formed with a first protrusion **13**, a second protrusion **14**, and a recess **15**. In FIG. **8**, the recess **15** is shown by a region of a plurality of points. The first protrusion **13** protrudes from the first main surface **111** of the pad portion **11**. The first protrusion **13** surrounds the semiconductor element **30** when viewed in the thickness direction z. The first protrusion **13** is located between the first region **111**A of the first main surface **111** and the second region **111**B of the first main surface **111**. The second protrusion **14** protrudes from the first main surface **111** in the same manner as the first protrusion **13**. The second protrusion **14** surrounds the first protrusion **13**. The second protrusion **14** is surrounded by the second region **111**B. The first protrusion **13** and the second protrusion **14** are in contact with the sealing resin **50**. The recess **15** is recessed from the first main surface **111**. The recess **15** surrounds the first protrusion **13** and is surrounded by the second protrusion **14**. Therefore, the recess **15** is located between the first protrusion **13** and the second protrusion **14**. The sealing resin **50** enters the recess **15**.

(31) The recess **15** is formed by subjecting the first main surface **111** to, for example, laser machining. At this time, an output of the laser with which the first main surface **111** is irradiated is set to be larger than an output of the laser with which the second region **111**B is irradiated when forming each of the plurality of grooves **16**. As a result, a depth D of the recess **15** from the first main surface **111** becomes greater than a depth d of each of the plurality of grooves **16** from the second region **111**B. With the formation of the recess **15**, the first base **101** of the pad portion **11** rises from the first main surface **111**. This rising is made along a direction in which the recess **15** extends. A portion raised from the first main surface **111** becomes the first protrusion **13** and the second protrusion **14**.

(32) As shown in FIGS. **3** and **6**, the plurality of terminals **20** are located apart from the support member **10** and are electrically connected to the semiconductor element **30**. As a result, the plurality of terminals **20** form a portion of the conductive path between the semiconductor element **30** and the wiring board on which the semiconductor device A**10** is mounted. The plurality of terminals **20** include the second base **201**. The second base **201** forms the main part of the plurality of terminals **20**. The second base **201** is also obtained from the lead frame from which the first base **101** of the support member **10** is obtained. Therefore, the second base **201** contains copper or a copper alloy.

(33) As shown in FIGS. **3** and **6**, the plurality of terminals **20** include a covering portion **21** and an exposed portion **22**. In the semiconductor device A**10**, the covering portion **21** and the exposed portion **22** include the second base **201**. The covering portion **21** is covered with the sealing resin **50**. The covering portion **21** includes a second main surface **211**. The second main surface **211** faces the same side as the first main surface **111** of the pad portion **11** of the support member **10** in the thickness direction z. The second main surface **211** is in contact with the sealing resin **50**. As shown in FIGS. **3** and **10**, a plurality of grooves **23** recessed from the second main surface **211** are formed in the covering portion **21**. The plurality of grooves **23** surround a portion of the second main surface **211** with which a second bonding layer **49** (details thereof will be described later) is in contact. The sealing resin **50** enters in the plurality of grooves **23**. The plurality of grooves **23** are formed by subjecting the second main surface **211** to laser machining, similarly to the plurality of grooves **16** formed in the pad portion **11**. The exposed portion **22** is connected to the covering portion **21** and is exposed from the sealing resin **50**. The exposed portion **22** extends from the covering portion **21** to the side away from the pad portion **11** of the support member **10** in the first direction x. The surface of the exposed portion **22** is plated with, for example, tin.

(34) As shown in FIG. **3**, in the semiconductor device A**10**, the plurality of terminals **20** include a first terminal **20**A and a second terminal **20**B. The first terminal **20**A extends along the first direction x and is located near the terminal portion **12** of the support member **10** in the second

direction y. The first terminal **20**A is electrically connected to the first electrode **31** of the semiconductor element **30**. Therefore, the first terminal **20**A corresponds to a source terminal of the semiconductor device A**10**.

(35) As shown in FIG. **3**, the second terminal **20**B extends along the first direction x and is located on the opposite side of the first terminal **20**A with the terminal portion **12** of the support member **10** interposed therebetween in the second direction y. The second terminal **20**B is electrically connected to the third electrode **33** of the semiconductor element **30**. Therefore, the second terminal **20**B corresponds to a gate terminal of the semiconductor device A**10**.

(36) As shown in FIG. **5**, in the semiconductor device A**10**, the portion exposed from the sealing resin **50** of the terminal portion **12** of the support member **10**, the exposed portion **22** of the first terminal **20**A, and the exposed portion **22** of the second terminal **20**B have the same height h. When viewed in the second direction y, a part of the terminal portion **12** overlaps the exposed portion **22** of the first terminal **20**A and the exposed portion **22** of the second terminal **20**B.

(37) As shown in FIG. **3**, the plurality of conductive members **40** are conductively bonded to the semiconductor element **30** and the plurality of terminals **20**. As a result, mutual conduction between the semiconductor element **30** and the plurality of terminals **20** is achieved. The plurality of conductive members **40** include a first member **41** and a second member **42**.

(38) As shown in FIGS. **3**, **9**, and **10**, the first member **41** is conductively bonded to the first electrode **31** of the semiconductor element **30** and the second main surface **211** of the covering portion **21** of the first terminal **20**A. As a result, the first terminal **20**A is electrically connected to the first electrode **31**. The composition of the first member **41** includes copper. In the semiconductor device A**10**, the first member **41** is a metal clip. The first member **41** is conductively bonded to the first electrode **31** and the second main surface **211** via the second bonding layer **49**. The second bonding layer **49** contains a metal element. The metal element is, for example, tin. The second bonding layer **49** is solder. As shown in FIG. **9**, a thickness t**2** of the second bonding layer **49** is smaller than a thickness t**1** of the first bonding layer **39**. In addition, the first member **41** may be a wire. In this case, since the first member **41** is formed by wire bonding, the second bonding layer **49** becomes unnecessary.

(39) As shown in FIG. **3**, the second member **42** is conductively bonded to the third electrode **33** of the semiconductor element **30** and the second main surface **211** of the covering portion **21** of the second terminal **20**B. As a result, the second terminal **20**B is electrically connected to the third electrode **33**. The second member **42** is a wire. The second member **42** is formed by wire bonding. The composition of the second member **42** includes aluminum (Al).

(40) The differences between the first member **41** and the second member **42** will be described below. Young's modulus (elastic modulus) of the second member **42** is smaller than Young's modulus of the first member **41**. This is based on the feature that, as described above, the composition of the first member **41** contains copper and the composition of the second member **42** contains aluminum. Therefore, a linear expansion coefficient of the second member **42** is greater than a linear expansion coefficient of the first member **41**. In addition, thermal conductivity of the second member **42** is smaller than thermal conductivity of the first member **41**. Further, as shown in FIG. **8**, a width B of the first member **41** is larger than a width (diameter) D of the second member **42**.

(41) As shown in FIGS. **6** and **7**, the sealing resin **50** covers the semiconductor element **30**, the plurality of conductive members **40**, and a portion of each of the support member **10** and the plurality of terminals **20**. The sealing resin **50** has electrical insulation. The sealing resin **50** is made of a material containing, for example, a black epoxy resin. The sealing resin **50** has a top surface **51**, a bottom surface **52**, a pair of first side surfaces **53**, a pair of second side surfaces **54**, a pair of openings **55**, and a mounting hole **56**.

(42) As shown in FIGS. **6** and **7**, the top surface **51** faces the same side as the first main surface **111** of the pad portion **11** of the support member **10** in the thickness direction z. As shown in FIGS. **5** to

7, the bottom surface **52** faces the opposite side of the top surface **51** in the thickness direction z. The back surface **112** of the pad portion **11** is exposed from the bottom surface **52**.

(43) As shown in FIGS. **2** and **4**, the pair of first side surfaces **53** are located apart from each other in the first direction x. The pair of first side surfaces **53** are connected to the top surface **51** and the bottom surface **52**. As shown in FIG. **5**, a portion of the terminal portion **12** of the support member **10** and the exposed portion **22** of the first terminal **20**A and the second terminal **20**B are exposed from one of the pair of first side surfaces **53**.

(44) As shown in FIGS. **2**, **4**, and **5**, the pair of second side surfaces **54** are located apart from each other in the second direction y. The pair of second side surfaces **54** are connected to the top surface **51** and the bottom surface **52**. As shown in FIG. **2**, the pair of openings **55** are located apart from each other in the second direction y. Each of the pair of openings **55** is recessed inward from the top surface **51** and one of the pair of second side surfaces **54** toward the sealing resin **50**. The first main surface **111** of the pad portion **11** of the support member **10** is exposed from the pair of openings **55**. As shown in FIGS. **2**, **4**, and **7**, the mounting hole **56** extends from the top surface **51** to the bottom surface **52** in the thickness direction z and penetrates the sealing resin **50**. When viewed in the thickness direction z, the mounting hole **56** is included in the through-hole **113** of the pad portion **11** of the support member **10**. An inner peripheral surface of the pad portion **11** that defines the through-hole **113** is covered with the sealing resin **50**. As a result, the maximum dimension of the mounting hole **56** is smaller than a dimension of the through-hole **113** when viewed in the thickness direction z.

Modification of the First Embodiment

(45) Next, a semiconductor device A**11**, which is a modification of the semiconductor device A**10**, will be described with reference to FIG. **11**. Here, the position of FIG. **11** is the same as the position of FIG. **8**. Similar to FIG. **8**, the sealing resin **50** is transparent in FIG. **11**. Further, in FIG. **11**, the second region **111**B of the first main surface **111** of the pad portion **11** (the support member **10**) is shown by a region of a plurality of straight lines, and the recess **15** formed in the pad portion **11** is shown by a region of a plurality of points.

(46) The semiconductor device A**11** is different from the semiconductor device A**10** in the configuration of the first main surface **111** of the pad portion **11** of the support member **10**. As shown in FIG. **11**, a distance (a distance in a direction orthogonal to the thickness direction z) from the second protrusion **14** to a boundary between the first region **111**A of the first main surface **111** and the second region **111**B of the first main surface **111** is longer than the corresponding distance in the semiconductor device A**10**. As a result, the boundary between the first region **111**A and the second region **111**B surrounds the second protrusion **14**.

(47) Next, operation and effects of the semiconductor device A**10** will be described.

(48) The semiconductor device A**10** includes the support member **10** including the main surface (the first main surface **111**), the semiconductor element **30** mounted on the main surface, and the bonding layer (the first bonding layer **39**) interposed between the support member **10** and the semiconductor element **30**. The support member **10** is formed with the first protrusion **13** protruding from the first main surface **111**. The first protrusion **13** surrounds the semiconductor element **30** when viewed in the thickness direction z. According to this configuration, in a case where the semiconductor element **30** is bonded to the support member **10** in the manufacture of the semiconductor device A**10**, when the first bonding layer **39** melted by reflow tries to wet and spread on the first main surface **111** in a direction orthogonal to the thickness direction z, the first bonding layer **39** comes into contact with the first protrusion **13**. As a result, the wetting and spreading of the first bonding layer **39** are restricted. Therefore, according to the semiconductor device A**10**, it is possible to suppress excessive wetting and spreading of the first bonding layer **39** interposed between the support member **10** and the semiconductor element **30**.

(49) The semiconductor device A**10** further includes the sealing resin **50** that covers at least a portion of the support member **10** and the semiconductor element **30**. The sealing resin **50** is in

contact with the first main surface **111** of the support member **10**. The first main surface **111** includes the first region **111**A that overlaps the semiconductor element **30** when viewed in the thickness direction z, and the second region **111**B having surface roughness larger than that of the first region **111**A. The second region **111**B surrounds the first region **111**A. The sealing resin **50** in contact with the second region **111**B exhibits an anchoring effect on the support member **10**. As a result, bonding strength of the sealing resin **50** with respect to the main surface **111** increases. Therefore, it is possible to improve adhesion between the support member **10** and the sealing resin **50**.

(50) Further, when the semiconductor device A**10** is used, a thermal strain is generated in the support member **10** due to heat generated from the semiconductor element **30**. As a result, a shear stress is generated at the interface between the first main surface **111** and the sealing resin **50**. When the adhesion between the support member **10** and the sealing resin **50** is improved, the shear stress reaching the first bonding layer **39** is reduced. Therefore, cracks generated in the first bonding layer **39** can be suppressed. When the cracks generated in the first bonding layer **39** are suppressed, it is possible to prevent the semiconductor element **30** from peeling off from the support member **10** and electrical resistance of the first bonding layer **39** from increasing (in a case where the first bonding layer **39** functions as a conductive member).

(51) The first protrusion **13** is located between the first region **111**A of the first main surface **111** and the second region **111**B of the first main surface **111**. As a result, since it is possible to prevent the first bonding layer **39** melted by reflow from reaching the second region **111**B, it is possible to prevent a decrease in the adhesion between the support member **10** and the sealing resin **50**.

(52) The support member **10** is formed with the second protrusion **14** that protrudes from the first main surface **111** and surrounds the first protrusion **13**. The second protrusion **14** is surrounded by the second region **111**B of the first main surface **111**. As a result, even in a case where the first bonding layer **39** melted by reflow overcomes the first protrusion **13**, the first bonding layer **39** comes into contact with the second protrusion **14**, such that the wetting and spreading of the first bonding layer **39** are doubly restricted. Therefore, it is possible to more reliably suppress the arrival of the first bonding layer **39** at the second region **111**B while effectively suppressing the excessive wetting and spreading of the first bonding layer **39**.

(53) The support member **10** is formed with the recess **15** that is recessed from the first main surface **111** and surrounds the first protrusion **13**. The recess **15** is surrounded by the second protrusion **14**. As a result, even in a case where the first bonding layer **39** melted by reflow overcomes the first protrusion **13**, the first bonding layer **39** enters the recess **15**. For this reason, it is difficult for the first bonding layer **39** to reach the second protrusion **14**. Therefore, it is possible to more effectively suppress the excessive wetting and spreading of the first bonding layer **39**.

(54) As the first protrusion **13** and the second protrusion **14** come into contact with the sealing resin **50**, the support member **10** exhibits an anchoring effect on the sealing resin **50**. Further, as the sealing resin **50** enters the recess **15**, the sealing resin **50** exhibits an anchoring effect on the support member **10**. As a result, the adhesion between the support member **10** and the sealing resin **50** is further improved.

(55) The semiconductor device A**10** further includes a conductive member **40** (the first member **41**) conductively bonded to the first electrode **31** of the semiconductor element **30** and the terminal **20** (the first terminal **20**A). The conductive member **40** is conductively bonded to the second main surface **211** of the terminal **20** via the second bonding layer **49**. A portion of the second main surface **211** to which the second bonding layer **49** is in contact is surrounded by the plurality of grooves **23** recessed from the second main surface **211**. As a result, when the sealing resin **50** comes into contact with the second main surface **211**, the sealing resin **50** exhibits an anchoring effect on the terminal **20**. For this reason, the adhesion between the terminal **20** and the sealing resin **50** is improved, and the shear stress generated at the interface between the second main surface **211** and the sealing resin **50** is less likely to reach the second bonding layer **49**. Therefore,

since generation of cracks in the second bonding layer **49** are suppressed, it is possible to prevent pitting corrosion from occurring in the conductive member **40**.
(56) The thickness t**1** of the first bonding layer **39** is larger than the thickness t**2** of the second bonding layer **49**. As a result, when the semiconductor device A**10** is used, heat generated from the semiconductor element **30** is likely to be conducted to the pad portion **11** of the support member **10** having a volume larger than that of each of the plurality of conductive members **40**. This makes it possible to improve heat dissipation of the semiconductor device A**10**.
(57) The composition of the first base **101** of the support member **10** includes copper. Further, the thickness T of the first base **101** of the pad portion **11** of the support member **10** is larger than the maximum thickness t.sub.max of the terminal **20**. As a result, it is possible to increase efficiency of heat conduction in the direction orthogonal to the thickness direction z while improving the heat conductivity of the pad portion **11**. This contributes to the improvement of the heat dissipation of the support member **10**.
(58) The pad portion **11** includes the back surface **112** facing the opposite side of the first main surface **111** in the thickness direction z. The back surface **112** is exposed from the bottom surface **52** of the sealing resin **50**. As a result, while the sealing resin **50** protects the semiconductor element **30** and the conductive member **40** from external factors, it is possible to avoid a decrease in heat dissipation of the semiconductor device A**10**.
Second Embodiment
(59) A semiconductor device A**20** according to a second embodiment of the present disclosure will be described with reference to FIGS. **12** to **16**. In these figures, the same or similar elements of the above-described semiconductor device A**10** are denoted by the same reference numerals, and explanation thereof will be omitted. Here, in FIGS. **12** and **14**, the sealing resin **50** is transparent for the sake of convenience of understanding. In FIG. **12**, the transparent sealing resin **50** is indicated by an imaginary line. In FIGS. **12** and **14**, the second region **111**B of the first main surface **111** of the pad portion **11** (the support member **10**) is shown by a region of a plurality of straight lines. In FIG. **14**, the recess **15** formed in the pad portion **11** is shown by a region of a plurality of points.
(60) The semiconductor device A**20** is different from the above-described semiconductor device A**10** in the configuration of the support member **10** and the plurality of terminals **20**.
(61) As shown in FIGS. **12** to **15**, the pad portion **11** of the support member **10** includes a first metal layer **102** in addition to the first base **101**. In this case, the first base **101** of the pad portion **11** includes the first main surface **111**. The first metal layer **102** is laminated on the first region **111**A of the first main surface **111** and is in contact with the first region **111**A. The thickness of the first metal layer **102** is smaller than the thickness of the first base **101**. The composition of the first metal layer **102** contains silver (Ag). In addition, the composition of the first metal layer **102** may contain nickel (Ni).
(62) As shown in FIG. **15**, the first bonding layer **39** is in contact with the first metal layer **102** of the pad portion **11** and the second electrode **32** of the semiconductor element **30**. As shown in FIG. **14**, the peripheral edge **102**A of the first metal layer **102** surrounds the semiconductor element **30** when viewed in the thickness direction z. The first protrusion **13** formed on the pad portion **11** and the second region **111**B of the first main surface **111** of the pad portion **11** surround the first metal layer **102**.
(63) As shown in FIGS. **12**, **13**, and **16**, the covering portion **21** of the plurality of terminals **20** includes a second metal layer **202** in addition to the second base **201**. In this case, the second base **201** of the covering portion **21** includes the second main surface **211**. The second metal layer **202** is laminated on the second main surface **211**. The thickness of the second metal layer **202** is smaller than the thickness of the second base **201**. The second metal layer **202** is surrounded by the plurality of grooves **23** formed in the covering portion **21**. When viewed in the thickness direction z, the area of the second metal layer **202** is smaller than the area of the first metal layer **102** of the pad portion **11** of the support member **10**. The composition of the second metal layer **202** includes

silver. In addition, the composition of the second metal layer **202** may include nickel.
(64) As shown in FIGS. **12**, **15**, and **16**, the first member **41** is conductively bonded to the first electrode **31** of the semiconductor element **30** and the second metal layer **202** of the covering portion **21** of the first terminal **20**A. The second bonding layer **49** is in contact with the second metal layer **202**. As shown in FIGS. **12** and **14**, the second member **42** is conductively bonded to the third electrode **33** of the semiconductor element **30** and the second metal layer **202** of the covering portion **21** of the second terminal **20**B.
Modification of Second Embodiment
(65) Next, a semiconductor device A**21**, which is a modification of the semiconductor device A**20**, will be described with reference to FIG. **17**. Here, the position of FIG. **17** is the same as the position of FIG. **15**.
(66) The semiconductor device A**21** is different from the semiconductor device A**20** in the configuration of the first metal layer **102** of the pad portion **11** of the support member **10**. The first metal layer **102** is in contact with the first region **111**A and the second region **111**B of the first main surface **111**. A plurality of slits **102**B leading to the plurality of grooves **16** are formed in the first metal layer **102** overlapping the second region **111**B. The sealing resin **50** enters the plurality of grooves **16** via the plurality of slits **102**B.
(67) Next, operation and effects of the semiconductor device A**20** will be described.
(68) The semiconductor device A**20** includes the support member **10** including the main surface (the first main surface **111**), the semiconductor element **30** mounted on the main surface, and the bonding layer (the first bonding layer **39**) interposed between the support member **10** and the semiconductor element **30**. The support member **10** is formed with the first protrusion **13** protruding from the main surface. The first protrusion **13** surrounds the semiconductor element **30** when viewed in the thickness direction z. Therefore, the semiconductor device A**20** also makes it possible to suppress excessive wetting and spreading of the bonding layer interposed between the support member **10** and the semiconductor element **30**.
(69) The support member **10** includes the base (the first base **101**) including the first main surface **111** and the metal layer (the first metal layer **102**) laminated on the first main surface **111**. The first bonding layer **39** is in contact with the first metal layer **102**. As a result, when the semiconductor element **30** is bonded to the support member **10** in the manufacture of the semiconductor device A**10**, since the wettability of the first bonding layer **39** with respect to the support member **10** is improved, the bonding strength of the semiconductor element **30** with respect to the support member **10** may be increased. In this case, the peripheral edge **102**A of the first metal layer **102** may surround the semiconductor element **30** when viewed in the thickness direction z. As a result, the bonding strength of the semiconductor element **30** with respect to the support member **10** may be increased more reliably.
(70) The second region **111**B of the first main surface **111** of the support member **10** and the first protrusion **13** formed on the support member **10** surround the first metal layer **102**. In the first metal layer **102**, the melted first bonding layer **39** is more likely to wet and spread than the first base **101**. Therefore, with this configuration, the improvement of the wettability of the first bonding layer **39** with respect to the support member **10** and the suppression of the excessive wetting and spreading of the first bonding layer **39** are compatible.
(71) Further, as the semiconductor device A**20** has the same configuration as the semiconductor device A**10**, the semiconductor device A**20** also exerts the operation and effects related to the configuration.
(72) The present disclosure is not limited to the above-described embodiments. The specific configuration of each part of the present disclosure may be freely changed in design.
(73) The technical configurations of the semiconductor device and the method for manufacturing the semiconductor device according to the present disclosure are described below.
(74) [Supplementary Note 1]

(75) A semiconductor device including: a support member including a main surface facing a thickness direction; a semiconductor element mounted on the main surface; and a bonding layer interposed between the support member and the semiconductor element, wherein the support member is formed with a first protrusion that protrudes from the main surface, and wherein the first protrusion surrounds the semiconductor element when viewed in the thickness direction.
[Supplementary Note 2]
(76) The semiconductor device of Supplementary Note 1, wherein the bonding layer contains a metal element.
(77) [Supplementary Note 3]
(78) The semiconductor device of Supplementary Note 2, wherein the metal element is tin.
(79) [Supplementary Note 4]
(80) The semiconductor device of Supplementary Note 2 or 3, further including a sealing resin that covers at least a portion of the support member and the semiconductor element, wherein the sealing resin is in contact with the main surface.
(81) [Supplementary Note 5]
(82) The semiconductor device of Supplementary Note 4, wherein the support member includes a back surface facing an opposite side of the main surface in the thickness direction, and wherein the back surface is exposed from the sealing resin.
(83) [Supplementary Note 6]
(84) The semiconductor device of Supplementary Note 4 or 5, wherein the main surface includes a first region that overlaps the semiconductor element when viewed in the thickness direction, and a second region that surrounds the first region, and wherein surface roughness of the second region is greater than surface roughness of the first region.
[Supplementary Note 7]
(85) The semiconductor device of Supplementary Note 6, wherein the support member is formed with a plurality of grooves recessed from the second region and located apart from each other.
(86) [Supplementary Note 8]
(87) The semiconductor device of Supplementary Note 6 or 7, wherein the first protrusion is located between the first region and the second region.
(88) [Supplementary Note 9]
(89) The semiconductor device of Supplementary Note 8, wherein the support member is formed with a second protrusion that protrudes from the main surface and surrounds the first protrusion, and wherein the second protrusion is surrounded by the second region.
[Supplementary Note 10]
(90) The semiconductor device of Supplementary Note 9, wherein the support member is formed with a recess that is recessed from the main surface and surrounds the first protrusion, and wherein the recess is surrounded by the second protrusion.
[Supplementary Note 11]
(91) The semiconductor device of any one of Supplementary Notes 6 to 10, wherein the support member includes a base including the main surface and a metal layer laminated on the main surface, and wherein the bonding layer is in contact with the metal layer.
[Supplementary Note 12]
(92) The semiconductor device of Supplementary Note 11, wherein a peripheral edge of the metal layer surrounds the semiconductor element when viewed in the thickness direction.
(93) [Supplementary Note 13]
(94) The semiconductor device of Supplementary Note 12, wherein the second region surrounds the metal layer.
(95) [Supplementary Note 14]
(96) The semiconductor device of Supplementary Note 13, wherein the first protrusion surrounds the metal layer.

(97) [Supplementary Note 15]
(98) The semiconductor device of any one of Supplementary Notes 4 to 14, further including a terminal located apart from the support member, wherein the semiconductor element includes a first electrode provided on a side which the main surface faces in the thickness direction, and wherein the terminal is electrically connected to the first electrode.
[Supplementary Note 16]
(99) The semiconductor device of Supplementary Note 15, further including a conductive member conductively bonded to the first electrode and the terminal, wherein the conductive member is covered with the sealing resin.
[Supplementary Note 17]
(100) The semiconductor device of Supplementary Note 16, wherein the semiconductor element includes a second electrode provided on an opposite side of the first electrode in the thickness direction, and wherein the second electrode is conductively bonded to the support member via the bonding layer.
(101) According to the present disclosure in some embodiments, it is possible to suppress excessive wetting and spreading of a bonding layer interposed between a support member and a semiconductor element.
(102) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

Claims

1. A semiconductor device comprising: a support member including a main surface facing a thickness direction; a semiconductor element mounted on the main surface; a bonding layer interposed between the support member and the semiconductor element; and a sealing resin that covers at least a portion of the support member and the semiconductor element, wherein the support member is formed with a first protrusion that protrudes from the main surface, wherein the first protrusion surrounds the semiconductor element when viewed in the thickness direction, wherein the sealing resin is in contact with the main surface, wherein the main surface includes a first region that overlaps the semiconductor element when viewed in the thickness direction, and a second region that surrounds the first region, and wherein surface roughness of the second region is greater than surface roughness of the first region.
2. The semiconductor device of claim 1, wherein the bonding layer contains a metal element.
3. The semiconductor device of claim 2, wherein the metal element is tin.
4. The semiconductor device of claim 1, wherein the support member includes a back surface facing an opposite side of the main surface in the thickness direction, and wherein the back surface is exposed from the sealing resin.
5. The semiconductor device of claim 1, wherein the support member is formed with a plurality of grooves recessed from the second region and located apart from each other.
6. The semiconductor device of claim 1, wherein the first protrusion is located between the first region and the second region.
7. The semiconductor device of claim 6, wherein the support member is formed with a second protrusion that protrudes from the main surface and surrounds the first protrusion, and wherein the second protrusion is surrounded by the second region.
8. The semiconductor device of claim 7, wherein the support member is formed with a recess that

is recessed from the main surface and surrounds the first protrusion, and wherein the recess is surrounded by the second protrusion.

9. The semiconductor device of claim 1, wherein the support member includes a base including the main surface and a metal layer laminated on the main surface, and wherein the bonding layer is in contact with the metal layer.

10. The semiconductor device of claim 9, wherein a peripheral edge of the metal layer surrounds the semiconductor element when viewed in the thickness direction.

11. The semiconductor device of claim 10, wherein the second region surrounds the metal layer.

12. The semiconductor device of claim 11, wherein the first protrusion surrounds the metal layer.

13. The semiconductor device of claim 1, further comprising a terminal located apart from the support member, wherein the semiconductor element includes a first electrode provided on a side which the main surface faces in the thickness direction, and wherein the terminal is electrically connected to the first electrode.

14. The semiconductor device of claim 13, further comprising a conductive member conductively bonded to the first electrode and the terminal, wherein the conductive member is covered with the sealing resin.

15. The semiconductor device of claim 14, wherein the semiconductor element includes a second electrode provided on an opposite side of the first electrode in the thickness direction, and wherein the second electrode is conductively bonded to the support member via the bonding layer.